THE HILBERT–KUNZ FUNCTION AND F-SIGNATURE FUNCTION OF THE DIAGONAL HYPERSURFACES

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Abstract. In this notes, I explain the algorithm to compute the Hilbert–Kunz function and the F-signature function of any given diagonal hypersurface, i.e., of the form $x_1^{d_1}+\cdots+x_n^{d_n}$, in any characteristic.

Computations. Some examples are followed by a short SageMath session. To reproduce them, start Sage in the folder containing the file `hilbert_kunz.py` (which comes with the accompanying notebooks) and type `load("hilbert_kunz.py")`. Lines starting with `sage:` are the input; the other lines are the output.

1. On the shape of Hilbert–Kunz functions and diagonal hypersurfaces

The Hilbert–Kunz function of a local ring is always of the following form [Mon83]: there is some $c \in \mathbb{R}_{\geq 1}$ such that

$$\mathrm{HK}_e(R) = cp^{de} + \Delta_e, \text{ where } \Delta_e = O(p^{e(d-1)}).$$

It would be great if $\mathrm{HK}_e(R)$ is eventually polynomial as its sister the Hilbert–Samuel function. However, already in the case of dimension 1, Monsky showed in that same article that Δ_e can be a periodic function:

Example 1.1. Let $R = \mathbb{F}_2[[x,y]]/(x^5+y^5)$, then for all $e \geq 0$,

$$\mathrm{HK}_e(R) = \begin{cases} 5\cdot 2^e - 4 & e \text{ even}, \\ 5\cdot 2^e - 6 & e \text{ odd}. \end{cases}$$

```
sage: S.<x,y> = GF(2)[]
sage: HK_function_diagonal(x^5 + y^5)
HK_e = 5 * 2^e - 4   for even e >= 0
HK_e = 5 * 2^e - 6   for odd e >= 1
```

The next obvious guess is that they might be "quasi-polynomials" on the variable p^e, i.e., polynomials $c_n(e)p^{en} + \ldots + c_1(e)p^e + c_0(e)$ whose coefficients $c_i(e)$ are periodic functions. This is also wrong, as we will see next, but here are several examples that might induce one to think that this is the case:

Examples 1.2. (1) Kunz provided the example [Kun76, Example 4.3], which is $R = \mathbb{F}_p[[t, x_1, \dots, x_n]]/(t^n - x_1^{d_1} \dots x_n^{d_n})$ for $p \neq |n \geq 2$. Then, the Hilbert–Kunz function is a quasi-polynomial. For instance, if $p = 5$, then for the hypersurface $t^3 - x^2y^2z^2$ we obtain

$$\mathrm{HK}_e(R) = \frac{26}{9} \cdot 5^{3e} - \frac{8}{3} \cdot 5^e + (-1)^e \frac{16}{9} \qquad e \geq 1.$$

One can compute this directly from Monsky's example, or with the algorithm in [CH98].

(2) This observation was told to me by Ilya Smirnov. Given a ring $(R, \mathfrak{m})$, the

$$\mathrm{HK}_e(R[[t]]) = p^e \mathrm{HK}_e(R).$$

This way, if you get a ring with quasi-polynomial Hilbert–Kunz function (like the two examples we have described so far), one can produce some trivial examples of higher-dimensional quasi-polynomial Hilbert–Kunz functions.

(3) Affine semigroup rings also give quasi-periodic Hilbert–Kunz functions.

In 1993, Han and Monsky [HM93] provided a setup and an algorithm to compute the Hilbert–Kunz function of the diagonal hypersurfaces, yielding interesting examples like the following:

Example 1.3. For all $e \geq 0$,

$$\mathrm{HK}_e\left(\frac{\mathbb{F}_5[x, y, z, t]}{(x^4 + y^4 + z^4 + t^4)}\right) = \frac{168}{61} 5^{3e} - \frac{107}{61} 3^e.$$

This example might suggest that Hilbert–Kunz functions are quasi-polynomial in several variables instead, like is the case above for the variables 5^e and 3^e. Again, this is in general not the case, as some nasty examples below will showcase.

Remark 1.4. One could argue that this is the case for all diagonal hypersurfaces, looking at the theorem above: define $e \mapsto k(e)$ where $0 \leq k(e) < \mu$ such that $e \equiv k(e) \pmod{\mu}$; we then have that

$$\ell^{\lfloor e/\mu \rfloor} = \ell^{\frac{e-k}{\mu}} = \ell^{-k(e)/\mu} \ell^{\frac{e}{\mu}},$$

which makes $\ell^{\lfloor e/\mu \rfloor}$ into a quasi-polynomial on the variable $\ell^{\frac{1}{\mu}}$. It is up to the reader to judge whether we want to allow irrational variables and hence be happy throwing diagonal hypersurfaces into the quasi-polynomial-in-several-variables drawer.

As already hinted above, Hilbert–Kunz functions might not even look like periodic-function combinations of powers. A few years after Han and Monsky's results, Chiang and Hung, building on these techniques, produced an algorithm to compute the Hilbert–Kunz function of any hypersurface of the form $\sum_{i=0}^{m} \prod_j^{n_j} x_{i,j}^{d_{i,j}}$

Example 1.5 (Example[CH98]). For every $e \geq 0$,

$$\mathrm{HK}_e\left(\frac{\mathbb{F}_7[[x,y,z,s,t,u,v]]}{((xyzs)^6+t^6+u^6+v^6)}\right) = \frac{2050747718345259}{429658164277013}\cdot 7^{6e} - \frac{267597}{74480}\cdot 7^{3e} + \left(\frac{164565}{2801}e - \frac{312186671451}{878707312}\right)7^e + \frac{104440051}{294115}\cdot 3^e.$$

Since (y,z,s,t,u,v) is a parameter ideal and minimal reduction of the maximal, and x^5 is a socle element, we can use

```
sage: S.<x,y,z,s,t,u,v> = GF(7)[]
sage: HK_function_disjoint_monomials((x*y*z*s)^6 + t^6 + u^6 + v^6)
HK_e = 2050747718345259/429658164277013 * 7^(6e)
       - 267597/74480 * 7^(3e)
       + (164565e/2801 - 312186671451/878707312) * 7^e
       + 104440051/294115 * 3^e
       for e >= 0
```

All these examples have in common that their Hilbert–Kunz series

$$\mathrm{HKS}(z) := \sum \mathrm{HK}_e(R) z^e,$$

are rational functions. The fact that a singularity has rational Hilbert–Kunz series implies having rational Hilbert–Kunz multiplicity, as one may deduce from the theory of generating functions. We have known for some years now, however, that irrational Hilbert–Kunz multiplicities exists [Bre13], and we have explicit conjectural examples of this fact that showcase the mysterious forms the Hilbert–Kunz function can attain:

Example 1.6. Let $H = x^3+y^3+xyz+uv \in \mathbb{F}_2[[x,y,z,u,v]]$, which is conjectured to have irrational Hilbert–Kunz multiplicity [Mon08]. In the observation after Corollary 2.7 in *op. cit.*, it is shown that

$$\mathrm{HKS}(z) = \frac{1}{(1-4z)(1-16z)} + \frac{2z(1-6z)(1+2z)^{1/2}}{(1-2z)^{3/2}(1-4z)(1-16z)}.$$

One may transform this into a closed form for the Hilbert–Kunz function, and in this case, Claude Opus suggests the following. Let

$$f_e(x) := \sum_{k=0}^{e-1} \binom{k}{\lfloor k/2 \rfloor} x^k,$$

$$f(x) := \lim_{e\to\infty} f_e(x) = \frac{1}{2x}\left(\sqrt{\frac{1+2x}{1-2x}} - 1\right), \quad |x| < \frac{1}{2}$$

Then,

$$\mathrm{HK}_e(R) = c_0(e)\,2^{4e} + c_1(e)\,2^{2e} + c_2(e)\,2^e,$$

where

$$c_0(e) = \frac{61}{42} + \frac{1}{16}\cdot\frac{5}{21}\, f_e\left(\tfrac{1}{16}\right) \xrightarrow[e\to\infty]{} \frac{4}{3} + \frac{5}{14\sqrt{7}},$$
$$c_1(e) = -\frac{1}{6} + \frac{1}{12}\, f_e\left(\tfrac{1}{4}\right) \xrightarrow[e\to\infty]{} \frac{\sqrt{3}-2}{6},$$
$$c_2(e) = -\frac{2}{7}(1 + f_e\left(\tfrac{1}{2}\right)) = -\frac{1}{7}\left(4e + 3 - (-1)^e\right)\binom{e}{\lfloor e/2\rfloor}\frac{1}{2^e} \sim -\frac{4}{7}\sqrt{\frac{2e}{\pi}} \xrightarrow[e\to\infty]{} -\infty,$$

1.1. **The diagonal hypersurfaces.** As we mentioned above, Han and Monsky showed the following result:

Theorem 1.7 ([HM93])**.** *Let p be a prime number and $(d_1,\dots,d_n)\in\mathbb{Z}_{\geq 2}$. Then,*

$$HK_e\left(\frac{\mathbb{F}_p[[x_1,\dots,x_n]]}{(x_1^{d_1}+\cdots+x_n^{d_n})}\right) = cp^{(n-1)e} + c_1(e)\ell^{\lfloor e/\mu\rfloor}, \text{ for all } e\geq e_0.$$

where e_0,ℓ,μ are non-negative integers (often $\ell\neq p$), $e\mapsto c_e$ is a periodic function of period μ, and $c\geq 1$ a rational number. Also, when $n\geq 3$, then $p^{(n-3)\mu}\geq\ell\geq 0$, and if $p=2$ or $n=2,3$, then $\ell\in\{0,1\}$.

The goal of these notes is explaining the tools that go into the proof of this fact, using it to effectively compute the Hilbert–Kunz function of a diagonal hypersurface, and give an idea of the general argument. The interest here relies on the fact that, even decades later, the machinery behind this example keeps providing insights of the Hilbert–Kunz function of hypersurfaces [CH98, MT04, MT06, Cas25, Men26b, PSSY26].

1.2. **Key point: proving a functional equation.** For the rest of the notes, set k be a field of characteristic p. The computation of the Hilbert–Kunz function of these hypersurfaces is about trying to understand the following function:

$$\left(\frac{a_1}{d_1},\dots,\frac{a_n}{d_n}\right)\overset{D_k}{\longmapsto}\frac{1}{d_1\dots d_n}\dim_k\left(\frac{k[x_1,\dots,x_n]}{(x_1^{a_1},\dots,x_n^{a_n},x_1^{d_1}+\cdots+x_n^{d_n})}\right). \tag{1}$$

For this, Han would define a particular function $\ell:\mathbb{Z}_{\geq 0}\to\mathbb{Z}$, and then one of the main theorem of her thesis is the following:

Theorem 1.8 (Han–Monsky, [Han92])**.** *Let $q=p^e$ for some $e\geq 0$, $r=(r_1,\dots,r_n)\in\mathbb{Z}_{\geq 1}^n$, $z=(z_1,\dots,z_n)\in[0,1]^n$ and $z^*=(1-z_1,\dots,z_n)$. Then,*

$$D(p^e(r+z)) = \begin{cases}\ell(r)D(qz)+q^{n-1}\phi_r(z) & \sum r_i \text{ even},\\ \ell(r)D(qz^*)+q^{n-1}\phi_r(z) & \sum r_i \text{ odd}.\end{cases}$$

where ϕ_r is a certain polynomial that depends on r and p.

In her thesis, Han showed that this functional equation holded provided that the ℓ function had a particular property, called being "p-multiplicative", and she proved it for $p>2$

and $n = 2, 3$ or $p = 2$ and $n \geq 1$. However, Monsky later developed the theory of the representation ring, which reinterpreted D_k and ℓ_k, and showed to show that ℓ is indeed p-multiplicative, which implies the functional equation in full generality. We nowadays have a better language for all this, and it equates to understanding the multiplication table of the representation ring, which we will talk about in the next section.

2. The Han–Monsky ring

2.1. **Motivation.** One may motivate this by trying to understand the function D_k:

(1) **The 2-variable case.** Say we want to compute $D(a, b)$ for positive integers $a \leq b$. Then,

$$D(a, b) = \dim_k \left(\frac{k[x, y]}{(x^a, y^b, x + y)} \right).$$

In this case, it is easy, because

$$\frac{k[x, y]}{(x^a, y^b, x + y)} \cong \frac{k[x]}{(x^a)},$$

and this has dimension a. So $D(a, b) = a$.

(2) **The 3-variable case.** If we want to do $D(a, b, c)$, then it is more subtle, because this is the dimension of

$$\frac{k[x, y, z]}{(x^a, y^b, z^c, x + y + z)} \cong \frac{k[x, y]}{(x^a, y^b, (x + y)^c)}.$$

For example, if a, b, c do not satisfy the triangular inequality, so say $a + b < c$, then $(x^a, y^b, (x + y)^c) = (x^a, y^b)$. If, instead, they do satisfy the triangular inequality, then it is more complicated to give a general form.

(3) **From a dimension to decomposing a $k[T]$-module.** There is another perspective to this, which is studying the k-vector space

$$M_e := \frac{k[x_1, \ldots, x_n]}{(x_1^{a_1}, \ldots, x_n^{a_n})}$$

as a $k[T]$-module where T acts like multiplication by $x_1^{d_1} + \cdots + x_n^{d_n}$. By the structure theorem of finitely generated modules over a PID, this module has some decomposition as a direct sum of cyclic $k[T]$-modules:

$$M_e = \oplus \left(\frac{k[T]}{(T^i)} \right)^{\oplus a_i}.$$

In other words, take the matrix describing the k-linear action of T in the k-vector space M_e over the monomial basis, compute its Jordan form, and count how many Jordan blocks of each size it has; these are the numbers a_i. Note that $D_k = \dim_k M_e/TM_e = \sum a_i$.

(4) **Decomposing as tensor product of simpler $k[T]$-modules.** With this perspective, let us now do a two-variable example:

Example 2.1. Say we want to compute $D_k(3/2, 4/3)$. If I have two $k[T]$-modules M, N, I can define a $k[T]$-module structure on the vector space $M \otimes_k N$ as $T(m \otimes n) = Tm \otimes n + m \otimes Tn$. With this (certainly arbitrary choice of a) product, I can decompose this vector space as a tensor product of simpler k-vector spaces:

$$M = \frac{k[x,y]}{(x^3, y^4)} = \frac{k[x]}{(x^3)} \otimes_k \frac{k[y]}{(y^4)},$$

and if we give the T-action given by x^2 and y^3, respectively, then this k-vector space isomorphism is actually also a $k[T]$-module isomorphism. I can compute the decomposition of each one of those simpler $k[T]$-modules, say:

$$M \cong \left(\frac{k[T]}{(T^2)} \oplus \frac{k[T]}{(T)}\right) \otimes_k \left(\frac{k[T]}{(T^2)} \oplus \left(\frac{k[T]}{(T)}\right)^{\oplus 2}\right). \tag{2}$$

So the problem becomes a problem about decomposing products of these simpler $k[T]$-modules. However, with a different notation we can get way further: we will work in the Grothendieck ring Γ_k of the isomorphism classes of these $k[T]$-modules.

Note that, in order to compute the Hilbert–Kunz function, it would be nice to have an easier description of the decomposition of M_e, as a function of e. All this setup leads to this, ultimately yielding a functional equation for D_k.

Definition 2.2. *(1)* ***k-objects.*** *a $k[T]$-module with nilpotent action of T, or equivalently, a $k[T]$-module that is finite-dimensional as k-vector space.*

(2) ***Product of k-objects.*** *We define their product as $T(m \otimes n) = Tm \otimes n + m \otimes Tn$.*

(3) ***Grothendieck ring of k-objects.*** *The relation $M - M' = N - N'$ if and only if $M \oplus N' \cong N \oplus M'$ is an equivalence class, and these operations distribute with the product of k-objects, so we can define Γ_k the Grothendieck ring of k-objects, whose unit element is $k[T]/(T) = k$.*

(4) ***δ-basis and λ-basis.*** *$\delta_i = k[T]/T^i$, and $\lambda_i := (-1)^i(\delta_{i+1} - \delta_i)$. Note that we set $\delta_0 = 0$, and $\delta_1 = \lambda_0$ is the unit.*

(5) ***Dimension and α-map.*** *Finally, we define $\dim_k(M - N) = \dim_k M - \dim_k N$ and $\alpha(M - N) := \dim_k(M/TM) - \dim_k(N/TN)$. Note that $\alpha(\delta_i) = 1$, $\alpha(\lambda_0) = 1$ and $\alpha(\lambda_i) = 0$ for $i \geq 1$.*

Example 2.3 (Back to the previous example with the δ-basis)**.** We continue with Example 2.1. Writing 2 in this language, we need to compute $(\delta_2 + \delta_1) \cdot (\delta_2 + 2\delta_1) = \delta_2^2 + 3\delta_1\delta_2 + 2\delta_1^2$, and we only need to compute the product δ_2^2—because δ_1 is the unit in Γ, the rest of the products are clear. If the characteristic is, say, $p = 2$, then, computing really is about explicitly writing what these are:

$$\delta_2^2 = \frac{k[T,S]}{(T^2, S^2)}$$

with T-action $T+S$. We can guess that $\delta_2^2 = 2\delta_2$ because $\delta_2^2 = a_1\delta_1 + a_2\delta_2$ for some a_1, a_2, since $(T+S)^2 = T^2 + S^2 = 0$, so there are no cyclic summands of larger order. However, since $\dim_k \delta_2^2 = 4$ and $\alpha\delta_2^2 = \dim_k k[T,S]/(T^2, S^2, T+S) = \dim_k k[T]/T^2 = 2$, we conclude that $\delta_2^2 = 2\delta_2$.

An alternative way of seeing this, noted to me by Kevin Tucker, is simply writing out the matrix representing the action of $T+S$ in a basis of monomials in $k[T,S]/(T^2,S^2)$, which is the matrix

$$M = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}, \qquad M^2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{pmatrix}.$$

Then, the number of Jordan blocks of size one will be $4 - 2\operatorname{rank} M + \operatorname{rank} M^2$, the number of Jordan blocks of size two is $\operatorname{rank} M - 2\operatorname{rank} M^2 + \operatorname{rank} M^3$.

```
sage: G = HanMonskyRing(2)
sage: G.delta(2)^2
2*delta_2
sage: M = (G.delta(2) + G.delta(1)) * (G.delta(2) + 2*G.delta(1))
sage: M
2*delta_1 + 5*delta_2
sage: M.alpha()    # alpha(M) = dim_k M/TM
7
sage: HanMonskyRing(3).delta(2)^2
delta_1 + delta_3
```

2.2. **The multiplication table of Γ_k.** In this section, we compile everything about the representation ring in the literature to answer the following question:

Question. *Given i, j, compute $\delta_i\delta_j$ or $\lambda_i\lambda_j$. In other words, what is the multiplication table of Γ_k in terms of the δ-basis and the λ-basis?*

One of the points of introducing the λ-basis is that its multiplication table is slightly easier to track, for several reasons. We will see later that the λ-basis also seemed to help answer a conjecture made by Han in her thesis [Han92], and this conjecture yielded the Hilbert–Kunz function of all diagonal hypersurfaces. Let us review some properties about the λ-basis to answer the question above.

Lemma 2.4 (Theorem 3.2, [HM93])**.** *The $\mathbb{Z}$-submodule $\Lambda_e \subset \Gamma$ generated by $\{\lambda_k\}_{k \le p^e - 1}$ is multiplicatively closed. In other words, if $\lambda_i\lambda_j = \sum a_k\lambda_k$ and $i, j \le p^e - 1$, then $a_k = 0$ for all $k \ge p^e$.*

Example 2.5 (Λ_1 in characteristic 2)**.** Now that we know this, let's compute the multiplication table of Λ_1 in characteristic 2. Obviously, since $\lambda_0 = \delta_1 = k[T]/(T)$ is the unit element

of the ring, then we know that $\lambda_0^2 = \lambda_0$ and $\lambda_0\lambda_1 = \lambda_1$. Let us do λ_1^2 then.

$$\lambda_1^2 = (\delta_2 - \delta_1)^2 = \delta_2^2 - 2\delta_2\delta_1 + \delta_1^2 = \delta_2^2 - 2\delta_2 + \delta_1.$$

Note that this implies that

$$\dim_k\left(\lambda_1^2\right) = \dim_k\left(\delta_2^2 - 2\delta_2 + \delta_1\right) = 1 \quad \text{and} \quad \alpha(\lambda_1^2) = 1,$$

because $\delta_2^2 = k[T,S]/(T^2, S^2, T+S)$ has dimension 2, so $\alpha(\lambda_1^2) = 2 - 2 + 1 = 1$. Therefore, since $\lambda_1^2 = a_0\lambda_0 + a_1\lambda_1$, it can only be that $a_0 = 1$ and $a_1 = 0$.

Actually, a similar argument shows that, in general, $\lambda_q^2 = 0$ in every characteristic.

Due to this fact, we may identify the elements of Λ_e with the matrices representing the action by multiplication over the λ-basis. We can start describing this representation for Λ_1 using Han and Monsky's formula:

Lemma 2.6 ([HM93])**.** *The multiplication table of Λ_1 is given as follows: for $0 \leq i \leq j \leq p-1$,*

$$\lambda_i\lambda_j = \sum_{k=j-i}^{\min(i+j,\, 2p-2-i-j)} \lambda_k.$$

Example 2.7 (Matrices for Λ_1 for $p = 2, 3, 5$)**.** For $p = 2$, the matrices are 2×2, and λ_0 is always the identity:

$$\lambda_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \lambda_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

For $p = 3$, the matrices are 3×3: λ_0 is the identity, λ_2 is the antidiagonal matrix, and λ_1 is a cross of 1's with 0's in the corners:

$$\lambda_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \lambda_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

For $p = 5$, the matrices are 5×5: λ_0 and λ_4 are the diagonal and antidiagonal, λ_2 is a filled diamond centered in the matrix, and λ_1, λ_3 are slanted diamonds (mirror images of each other):

$$\lambda_0 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \lambda_1 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix},$$

$$\lambda_2 = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \lambda_3 = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{pmatrix}, \qquad \lambda_4 = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Note that all these matrices are symmetric and commute, and therefore are simultaneously diagonalisable, which would yield an orthogonal basis of elements of Λ_e. This perspective is exploited in [CR26, Men26a] to understand the Hilbert–Kunz multiplicities of quadrics better. The first reference in particular focuses more on the matrix representation, following the approach of [PSSY26], which originates from this matrix representation, together with the Han–Monsky algorithm.

```
sage: G = HanMonskyRing(5)
sage: G.lam(1).matrix()    # column j is lambda_1 * lambda_j
[0 1 0 0 0]
[1 1 1 0 0]
[0 1 1 1 0]
[0 0 1 1 1]
[0 0 0 1 0]
sage: G.lam(1)^2
lambda_0 + lambda_1 + lambda_2
```

Lemma 2.8. *Let $i \geq 0$, and let $i_0 + i_1 p + i_2 p^2 + \ldots + i_{e-1} p^{e-1}$ its p-adic expansion. Then, $\lambda_i = \lambda_{i_0} \lambda_{i_1 p} \ldots \lambda_{i_{e-1} p^{e-1}}$. In particular, if I want to know $\lambda_i \lambda_j$, I only need to care about the products $\lambda_{aq} \lambda_{bq}$.*

Example 2.9. With this lemma, and the observation that $\lambda_q^2 = 0$, we can now give a complete description of multiplication in the Han–Monsky ring in characteristic 2: indeed, to compute $\lambda_a \lambda_b$, first decompose a and b in their 2-adic decompositions $a_0 + a_1 2 + \cdots + a_e 2^e$ and $b_0 + b_1 2 + \cdots + b_e 2^e$, where $a_k, b_k \in \{0, 1\}$; then $\lambda_a \lambda_b = \prod \lambda_{a_k 2^k} \lambda_{b_k 2^k}$. This product can be described with the concept of "nim sum": the nimsum of a tuple of zeros and ones is the component-wise sum modulo 2, denoted like

$$(a_0, ..., a_e) \oplus (b_0, ..., b_e) = (a_0 + b_0 \pmod 2, ..., a_e + b_e \pmod 2).$$

Thus, for instance,

$$\lambda_3 \lambda_5 = (\lambda_{0 \cdot 4} \lambda_{1 \cdot 2} \lambda_1) (\lambda_{1 \cdot 4} \lambda_{0 \cdot 2} \lambda_1) = \lambda_4 \lambda_2 \lambda_0 = \lambda_6.$$

In terms of matrices, in characteristic 2 it is easy to observe what is going on. Let us denote $[\lambda_i]_2$ for the matrix representing the product by λ_i in Λ_2, and $\otimes$ the Kronecker product of matrices:

$$[\lambda_1] = I_2 \otimes J = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad [\lambda_2] = J \otimes I_2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},$$

$$[\lambda_3] = [\lambda_1][\lambda_2] = J \otimes J = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad [\lambda_0] = I_4.$$

The issue when trying to scale these observations to higher characteristics is that the Kronecker product description does not work exactly.

Example 2.10. In characteristic 3, note that λ_3 has the following matrix in Λ_2:

$$[\lambda_3]_2 = \left(\begin{array}{ccc|ccc|ccc} 0&0&0&1&0&0&0&0&0\\ 0&0&0&0&1&0&0&0&0\\ 0&0&0&0&0&1&0&0&0\\ \hline 1&0&0&0&0&1&1&0&0\\ 0&1&0&0&1&0&0&1&0\\ 0&0&1&1&0&0&0&0&1\\ \hline 0&0&0&1&0&0&0&0&0\\ 0&0&0&0&1&0&0&0&0\\ 0&0&0&0&0&1&0&0&0 \end{array}\right),$$

which is not exactly $[\lambda_1]_1 \otimes [\lambda_0]_1$. One needs to adjust for some "reflections".

```
sage: G = HanMonskyRing(3)
sage: A = G.lam(3).matrix()
sage: B = G.lam(1).matrix().tensor_product(identity_matrix(3))
sage: A.subdivide([3,6], [3,6])    # 3x3 blocks, as above
sage: A, B    # left: [lambda_3]_2,  right: [lambda_1]_1 (x) [lambda_0]_1
(
[0 0 0|1 0 0|0 0 0]  [0 0 0|1 0 0|0 0 0]
[0 0 0|0 1 0|0 0 0]  [0 0 0|0 1 0|0 0 0]
[0 0 0|0 0 1|0 0 0]  [0 0 0|0 0 1|0 0 0]
[-----+-----+-----]  [-----+-----+-----]
[1 0 0|0 0 1|1 0 0]  [1 0 0|1 0 0|1 0 0]
[0 1 0|0 1 0|0 1 0]  [0 1 0|0 1 0|0 1 0]
[0 0 1|1 0 0|0 0 1]  [0 0 1|0 0 1|0 0 1]
[-----+-----+-----]  [-----+-----+-----]
[0 0 0|1 0 0|0 0 0]  [0 0 0|1 0 0|0 0 0]
[0 0 0|0 1 0|0 0 0]  [0 0 0|0 1 0|0 0 0]
[0 0 0|0 0 1|0 0 0], [0 0 0|0 0 1|0 0 0]
)
sage: A == B
False
```

The main result is the following, we refer the reader to [MT06] for the details:

Theorem 2.11 (Monsky and Teixeira, [MT06], Meng, [Men26a])**.** *Define the following map*

$$\theta(\lambda_i) := \begin{cases} \lambda_{pi} & \text{even } i, \\ \lambda_{pi+p-1} & \text{odd } i. \end{cases}$$

The $\mathbb{Z}$-linear extension to Γ_k of this map is a ring homomorphism.

This means that once we understand the multiplication table of Λ_1, then we understand the multiplication table of Γ_k: denote the p-adic expansion of i by $i_0 + i_1 p + \ldots + i_{e-1}p^{e-1}$;

Proposition 2.12 (Monsky–Teixeira, [MT06], Meng, [Men26a]). *For $i = \sum_{k=0}^{e-1} i_k p^k$ with $0 \le i_k \le p-1$, put*

$$a_k(i) := \begin{cases} i_k, & \sum_{r=k+1}^{e-1} i_r p^r \text{ even}, \\ p-1-i_k, & \sum_{r=k+1}^{e-1} i_r p^r \text{ odd}. \end{cases}$$

Then the $\mathbb{Z}$-linear map

$$\Phi_e \colon \Lambda_e \longrightarrow \Lambda_1^{\otimes e}, \qquad \lambda_i \longmapsto \lambda_{a_0(i)} \otimes \cdots \otimes \lambda_{a_{e-1}(i)}$$

is a ring isomorphism.

3. The Han–Monsky algorithm

We now come back to computing the Hilbert–Kunz function of a diagonal hypersurface. First, a useful extension of Γ:

Definition 3.1. *For a real number $t \ge 0$ write $t = n + \vartheta$ with $n = \lfloor t \rfloor$ and $\vartheta \in [0,1)$, and set*

$$\delta_t := (1-\vartheta)\,\delta_n + \vartheta\,\delta_{n+1} \ \in \ \Gamma_k \otimes_{\mathbb{Z}} \mathbb{R} \qquad (\delta_0 = 0),$$

so that $t \mapsto \delta_t$ is the linear interpolation of $i \mapsto \delta_i$. Extend α $\mathbb{R}$-linearly. Then, for $t_1, \ldots, t_n \in \mathbb{Q}$ put $D_k(t_1, \ldots, t_n) := \alpha\big(\prod_{i=1}^n \delta_{t_i}\big)$, and for integers $r_1, \ldots, r_n \ge 0$ put $\ell_k(r_1, \ldots, r_n) := \alpha\big(\prod_{i=1}^n \lambda_{r_i}\big)$.

We can write out observations at the beginning as:

Lemma 3.2 (Lemma 5.6, [HM93]). *If $a_1, \ldots, a_n, d_1, \ldots, d_n$ are non-negative integers, then*

$$\dim_k \left(\frac{k[x_1, \ldots, x_n]}{(x_1^{a_1}, \ldots, x_n^{a_n}, x_1^{d_1} + \cdots + x_n^{d_n})} \right) = d_1 \ldots d_n D_k(a_1/d_1, \ldots, a_n/d_n).$$

A corollary of Monsky and Teixeira's result above is a functional equation for ℓ_k:

Theorem 3.3 (The ℓ function is p-multiplicative, Theorem 4.5, [HM93]). *Let $q = p^e$ for some $e \ge 0$, let $k_1, \ldots, k_n < q$ and $r_1, \ldots, r_n \ge 0$. Then,*

$$\ell_k(qr + k) = \begin{cases} \ell(r)\ell(k_1, \ldots, k_n) & \text{if } \sum r_i \text{ even}, \\ \ell(r)\ell(q-1-k_1, k_2, \ldots, k_n) & \text{if } \sum r_i \text{ odd}. \end{cases}$$

To compute Hilbert–Kunz functions of diagonal hypersurfaces, we need to compute D_k, and the point is that we can also get a (more involved) functional equation for D_k:

Theorem 3.4 (Theorem 4.13, [HM93]). *Let $q = p^e$ for some $e \geq 0$, $u = (u_1, \dots, u_n) \in [0, q)^n$, and $r = (r_1, \dots, r_n) \in \mathbb{Z}^n_{\geq 1}$.*

$$D(qr + u) = \begin{cases} \ell(r)D(u_1, \dots, u_n) + q^{n-1}\phi_r\left(\frac{u}{q}\right) & \sum r_i \text{ even}, \\ \ell(r)D(q - u_1, u_2, \dots, u_n) + q^{n-1}\phi_r\left(\frac{u}{q}\right) & \sum r_i \text{ odd}, \end{cases}$$

where $\phi_r \in \mathbb{Z}[x_1, \dots, x_n]$ is the polynomial

$$\phi_r(x) = \begin{cases} \sum_{A \subsetneq \{1,\dots,n\}} n_A x_A & \sum r_i \text{ even} \\ \sum_{A \subsetneq \{1,\dots,n\}} n_A x_A - \ell(r_1, \dots, r_n) \prod_{i=2}^n x_i & \sum r_i \text{ odd} \end{cases}$$

for $n_A := \alpha\left(\left(\prod_{i \in A}(-1)^{r_i}\lambda_{r_i}\right)\left(\prod_{j \notin A}\delta_{r_j}\right)\right)$ and $x_A := \prod_{i \in A} x_i$. In other words, if $z = (z_1, \dots, z_n) \in [0, 1]^n$ and $z^* = (1 - z_1, \dots, z_n)$,

$$D(p^e(r + z)) = \begin{cases} \ell(r)D(qz) + q^{n-1}\phi_r(z) & \sum r_i \text{ even}, \\ \ell(r)D(qz^*) + q^{n-1}\phi_r(z) & \sum r_i \text{ odd}. \end{cases}$$

The setup when you give me a diagonal hypersurface is the following: let $\beta \in \mathbb{Q}^n_{\geq 0}$, and $p, \mu > 0$ (μ has to be found!). We define the following data:

(1) $r := \lfloor \beta \rfloor$ and $z := \beta - r \in [0, 1]^s$.

(2) $R := \lfloor p^\mu \beta \rfloor$ and $z' := p^\mu \beta - R \in [0, 1]^s$.

(3) $t = R - p^\mu r \in \mathbb{Z}^s$.

Note that $\ell(R) = \ell^\sharp \ell(r)$ where if $\sum r_i$ even, $\ell^\sharp := \ell(t)$, and if $\sum r_i$ odd, then $\ell^\sharp := \ell(p^\mu - 1 - t_1, t_2, ..., t_n)$.

Theorem 3.5 (Theorem 5.2). *If $z' = z$ and $\sum R_i \equiv \sum r_i \pmod 2$, then*

$$\frac{1}{p^{e(n-1)}}(D(p^{e+\mu}\beta) - \ell^\sharp D(p^e \beta))$$

is constant over $e \geq 0$.

Proof. This proof is the key that explains why the algorithm works. Let $q = p^e$, and let us start assuming $\sum R_i \equiv \sum r_i \equiv 0 \pmod 2$. By the previous result, and using that $z' = z$, we that

$$\begin{aligned} D(p^{e+\mu}\beta) - \ell^\sharp D(p^e \beta) &= D(q\,(R + z)) - \ell^\sharp D(q\,(r + z)) \\ &= \ell(R)D(qz) + q^{n-1}\phi_R(z) - \ell^\sharp\left(\ell(r)D(qz) + q^{n-1}\phi_r(z)\right) \\ &= q^{n-1}\left(\phi_R(z) - \ell^\sharp \phi_r(z)\right). \end{aligned}$$

Given that $\sum R_i$ and $\sum r_i$ have the same parity, then the term $\phi_R(z) - \ell^\sharp \phi_r(z)$ does not depend on e—note that the variables R, r, z and t only depend on p, β and μ. The same proof follows for theo ther parity case, noting the difference in the definitions of ϕ_r and $\ell^\sharp$. □

For any e, there need not be such μ, but if e is large enough, there will. We will take this into account.

Corollary 3.6. *Let $p > 0$ and $d_i \geq 2$ for $i = 1, \ldots, n$. Then,*

$$\frac{\mathbb{F}_p[[x_1, \ldots, x_n]]}{(x_1^{d_1} + \cdots + x_n^{d_n})}$$

and

$$I = (x_1^{a_1}, \ldots, x_n^{a_n}).$$

Set $\beta = \left(\frac{a_1}{d_1}, \ldots, \frac{a_n}{d_n}\right) \in \mathbb{Q}_{>0}$, and assume that there are some integers $e_0 \geq 0$ and $\mu \geq 1$ such that $z' = z$ and $\sum R_i \equiv \sum r_i \pmod 2$. Then, for every $e \geq e_0$,

$$HK_e\left(\frac{\mathbb{F}_p[[x_1, \ldots, x_n]]}{(x_1^{d_1} + \cdots + x_n^{d_n})}, (x_1^{a_1}, \ldots, x_n^{a_n})\right) = Cp^{(n-1)e} + \Delta_e$$

such that C is an integer, and $\Delta_{e+\mu} = \ell^\sharp \Delta_e$.

Proof. Recall that $\mathrm{HK}_e(R, I) = d_1 \ldots d_n D(p^e \beta)$. Say $d = d_1 \ldots d_n$. The previous theorem says that, for some constant A, for every $e \geq e_0$ we have that

$$\frac{1}{d}\mathrm{HK}_{e+\mu}(R, I) - \ell^\sharp \frac{1}{d}\mathrm{HK}_e(R, I) = \tilde{A} p^{e(n-1)},$$

or, equivalently, setting $A = \tilde{A} d$,

$$\mathrm{HK}_{e+\mu}(R, I) - \ell^\sharp \mathrm{HK}_e(R, I) = A p^{e(n-1)}.$$

We also know by Monsky that $\mathrm{HK}_e(R, I) = Cp^{e(n-1)} + \Delta_e$ for some function $\Delta_e = O(p^{e(n-2)})$. If we plug this into the equation above, and divide by $p^{e(n-1)}$, we obtain that for all $e \geq e_0$,

$$\frac{1}{p^{e(n-1)}}\left(\Delta_{e+\mu} - \ell^\sharp \Delta_e\right) = A + \ell^\sharp C - Cp^{\mu(n-1)}.$$

Since the left side converges to 0 when $e \to \infty$, but the right side does not depend on e, we conclude that $\Delta_{e+\mu} = \ell^\sharp \Delta_e$, and the result follows. □

3.1. **How to find e_0, μ and ℓ: number theoretical input.** One can always find such numbers making Corollary 3.6 work. Given any tuple of positive rationals

$$\tilde{\beta} := \left(\frac{a_1}{d_1}, \ldots, \frac{a_n}{d_n}\right).$$

with $a_i, d_i \geq 1$, we prepare some data:

(1) $e_0 := \max_i\{s : p^s \mid d_i \text{ for some } i\}$,

(2) $\mu = \min\{p^{k+e_0} \equiv 1 \pmod{d_i} \text{ for all } i\}$.

Put $\beta := p^{e_0}\tilde{\beta}$. It is then easy to check the following:

Lemma 3.7. *With r, z, R, z' defined from β as in the setup above:*

(a) *for every multiple* μ' *of* μ *we have* $z' = z$ *and* $R = r + (p^{\mu'} - 1)\beta$;

(b) *if* p *is odd and* $\mu = 2\mu_0$, *then moreover* $R_i \equiv r_i \pmod 2$ *for every* i, *so* $\sum R_i \equiv \sum r_i \pmod 2$;

(c) *if* $p = 2$, *then for every multiple* μ *of* μ_0, $\sum_i R_i - \sum_i r_i \equiv \sum_i b_i \pmod 2$.

In particular, you can apply 3.5.

4. F-signature function of diagonal hypersurfaces via Han–Monsky

This is one way of approaching this computation probably complementary to the techniques in [CSTZ24]. Both methods probably achieve different aspects of this, given that in *op. cit.*, some formulas are attained which I am not sure can be done directly through Han–Monsky.

Proposition 4.1 (Huneke-Leuschke, [HL04])**.** *Let* $(R, \mathfrak{m}, k)$ *be an F-finite Gorenstein local ring of dimension* d *and characteristic* p, *with* $k = k^p$. *Let* I *be an ideal generated by a system of parameters, and* $u \in R$ *be a socle element of* I, *so that* $(I : \mathfrak{m}) = I + (u)$. *Then, with* $q = p^e$,

$$\mathrm{FS}_e(R) = \ell\big(R/I^{[q]}\big) - \ell\big(R/(I + (u))^{[q]}\big) = \ell(R/I)\, q^d - \mathrm{HK}_e\left(R, I + (u)\right).$$

If I *is a minimal reduction of* $\mathfrak{m}$, *then* $\ell(R/I) = e(I) = e(R)$, *and so*

$$\mathrm{FS}_e(R) + \mathrm{HK}_e(R, I + (u)) = e(R)q^d.$$

Lemma 4.2. *Let* $f = x_1^{d_1} + \cdots + x_n^{d_n}$ *with* $n \geq 2$, $d_i \geq 2$, $R = k[[x_1, \ldots, x_n]]/(f)$ *and* $I = (x_1, \ldots, x_{n-1})R$, *and* $u = x_n^{d_n - 1}$ *generates the socle of* R/I. *Let* $J = (x_1, \ldots, x_{n-1}, x_n^{d_n-1})$. *Hence, for* k *perfect,*

$$FS_e(R) = d_n p^{e(n-1)} - HK_e(R, J).$$

In particular, for the F-signature function of every diagonal hypersurface we also conclude that

$$FS_e\left(\frac{\mathbb{F}_p[[x_1, \ldots, x_n]]}{(x_1^{d_1} + \cdots + x_n^{d_n})}\right) = cp^{(n-1)e} + c_1(e)\ell^{\lfloor e/\mu \rfloor}, \text{ for all } e \geq e_0.$$

where e_0, ℓ, μ *are non-negative integers (often* $\ell \neq p$), $e \mapsto c_e$ *is a periodic function of period* μ, *and* $c \geq 1$ *a rational number. Also, when* $n \geq 3$, *then* $p^{(n-3)\mu} \geq \ell \geq 0$, *and if* $p = 2$ *or* $n = 2, 3$, *then* $\ell \in \{0, 1\}$.

The F-signature function can then be computed by the Han–Monsky algorithm setting setting $\beta = (1/d_1, \ldots, 1/d_{n-1}, (d_n - 1)/d_n)$. Moreover $e(R) = \min_i d_i$, and I is a minimal reduction of $\mathfrak{m}$ if and only if $d_n = \min_i d_i$.

Example 4.3. Let $f = x^3 + y^3 + z^3 \in k[[x, y, z]]$ where k is perfect of characteristic $p > 0$. It follows from Fedder's criterion that if the characteristic $p \equiv 1 \pmod 3$, then $k[[x, y, z]]/(f)$ is F-pure, and if $p \equiv 2 \pmod 3$, then it is not. In the second case, $\mathrm{FS}_e(R) = 0$, and then

$\mathrm{HK}_e(R) = 3p^{2e}$, so let us focus on the first case. We will show here, via the Han–Monsky algorithm, that $\mathrm{FS}_e(f) = 1$ in the first case.

Let $J = (x, y, z^2)$. Then, by our observation above, to compute the F-signature function of this, we only need to apply Han and Monsky's algorithm for the ideal J.

Let $p \equiv 1 \pmod 3$, and set $a = \frac{p-1}{3}$. Then, one can conclude from Theorem can choose $e_0 = 0$ and $\mu = 1$ in Corollary 3.6: indeed, if we let $\beta = (\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$, then the integer part is $r = 0$, and $p\beta = (a, a, 2a) + \beta$. Thus, $z' = z$, $R = (a, a, 2a)$ and $\sum R_i = 4a \equiv 0 \equiv \sum r_i$, or in other words, the hypotheses of said corollary are verified. Since the example is two-dimensional, then $\ell = 0$ or 1, and the Hilbert–Kunz function at J will be of the form

$$\mathrm{HK}_e(R, J) = cp^{2e} - \Delta,$$

for c, Δ rational constants for all $e \geq 0$. Note that $2 = \mathrm{HK}_0(R, J) = c - \Delta_0$, so $\mathrm{HK}_1(R, J) = cp^2 - \Delta_0 = cp^2 + 2 - c = c(p^2 - 1) + 2$, and we conclude that

$$e_{\mathrm{HK}}(R, J) = \frac{\mathrm{HK}_1(R, J) - 2}{p^2 - 1}.$$

To compute $\mathrm{HK}_1(R, J)$ in general, we may use the techniques in the previous section (although, for a fixed characteristic p, one may compute this value with a computer). As $\frac{p}{3} = a + \frac{1}{3}$ and $\frac{2p}{3} = 2a + \frac{2}{3}$, then

$$\mathrm{HK}_1(R, J) = 27\,\alpha\big(\delta_{p/3}^2\,\delta_{2p/3}\big) = \alpha\big((2\delta_a + \delta_{a+1})^2(\delta_{2a} + 2\delta_{2a+1})\big),$$

and one can check that then $\mathrm{HK}_1(R, J) = 3p^2 - 1$.[1] Therefore,

$$e_{\mathrm{HK}}(R, J) = 3, \quad \Delta_0 = 1 \Rightarrow \mathrm{HK}_e(R, J) = 3p^{de} - 1,$$

and now using the result above, it follows $\mathrm{FS}_e(R) = 1$ for all $e \geq 1$. This is a well-known example of a hypersurface that is F-split but not strongly F-regular.

```
sage: S.<x,y,z> = GF(7)[]
sage: FS_function_diagonal(x^3 + y^3 + z^3)
FS_e = 1   for e >= 0
sage: S.<x,y,z> = GF(5)[]
sage: FS_function_diagonal(x^3 + y^3 + z^3)
FS_e = 0   for e >= 1
```

Question. *Can this example be scaled up to every dimension?*

[1] In general, one can often get to a general formula over p computing this with the matrix representation introduced before. However, Claude computed it like this: in Section 2.1 (2), we saw that $\alpha(\delta_i\delta_j\delta_l) = \dim_k k[x, y]/(x^i, y^j, (x + y)^l)$. If $l \geq i + j - 1$ then $(x + y)^l \in (x^i, y^j)$ and the value is ij; this covers all the triples needed except $(a + 1, a + 1, 2a)$. In $k[x, y]/(x^{a+1}, y^{a+1})$ the top degree $2a$ is spanned by $x^a y^a$ and $(x+y)^{2a} \equiv \binom{2a}{a} x^a y^a \pmod{(x^{a+1}, y^{a+1})}$; since $2a < p$, $\binom{2a}{a} \not\equiv 0 \pmod p$, so the ideal generated by $(x+y)^{2a}$ is $k\,x^a y^a$ and the value is $(a + 1)^2 - 1$. Therefore

$$\mathrm{HK}_1(R, J) = 4 \cdot 3a^2 + 4 \cdot 3a(a + 1) + \big(((a + 1)^2 - 1) + 2(a + 1)^2\big) = 27a^2 + 18a + 2 = 3p^2 - 1.$$

The issue is the computation of HK_1 and ℓ. In the next section, we briefly discuss how we could go about this question.

5. Using the matrix representation

As I explained above, one can express the elements of the representation ring, or more especifically, elements of the subrings Λ_e, as $p^e \times p^e$ matrices over the integers.

The results we showed on the previous section yield that the Hilbert–Kunz function can be computed from a few values of the function, like it is the case in the example of $x^3+y^3+z^3$, where it is totally recovered from just $\mathrm{HK}_0(R)$ and $\mathrm{HK}_1(R)$. However, computing $\mathrm{HK}_1(R)$ in a characteristic-free way requires a potentially subtle computation in Λ_1. It is in this situation that the matrix representation is very useful.

Example 5.1 ([Yos09, PSSY26])**.** Let $f = x_1^2 + \cdots + x_{d+1}^2$. Then, for every characteristic $p > 2$, the Hilbert–Kunz function for every $e \geq 0$ is $\mathrm{HK}_e(R) = cp^{de} + \Delta_e$ with $\Delta_{e+1} = \ell\Delta_e$, where $\ell = \alpha(\lambda_{\frac{p-1}{2}}^{d+1})$. Similarly, $\mathrm{HK}_1(R) = \alpha((\delta_{\frac{p-1}{2}}+\delta_{\frac{p+1}{2}})^{d+1})$. We may represent the k-objects $\lambda_{\frac{p-1}{2}}$ and $\delta_{\frac{p-1}{2}} + \delta_{\frac{p+1}{2}}$ by the following matrices: for instance, for $p = 5$ ($m = 2$):

$$[\lambda_2] = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}, \qquad [\delta_2 + \delta_3] = \begin{pmatrix} 2 & -2 & 1 & 0 & 0 \\ -2 & 1 & -1 & 1 & 0 \\ 1 & -1 & 1 & -1 & 1 \\ 0 & 1 & -1 & 1 & -2 \\ 0 & 0 & 1 & -2 & 2 \end{pmatrix}.$$

and therefore one may realize that the problem becomes computing the $(1,1)$-element of the $(d+1)$-th power of these matrices. Actually, one can proof that to solve this problem, one may as well consider the matrix of $[\delta_2 + \delta_3]$ without the minus signs.

In this particular case of the A_1 singularity, it is shown in [PSSY26] that this problem can be interpreted with Ehrhart theory. They actually recover the $\sec x + \tan x$ formula by these means, as well as showing that $p \mapsto e_{\mathrm{HK}}(R_{p,d})$, where $R_{p,d}$ is the coordinate ring of f over $\mathbb{F}_p$, is a rational function $f(p^2)/g(p^2)$ where $\deg f(p^2), \deg g(p^2) \leq d$.

```
sage: G = HanMonskyRing(5)
sage: L = G.lam(2).matrix()
sage: M = (G.delta(2) + G.delta(3)).matrix()
sage: (L^4)[0,0], (M^4)[0,0]    # d = 3: l and HK_1(R); [0,0] is the (1,1) entry
(5, 165)
```

References

[Bre13] Holger Brenner. Irrational Hilbert-Kunz multiplicities, 2013. arXiv:1305.5873.

[Cas25] Joel Castillo-Rey. Strong Watanabe-Yoshida conjecture for Complete Intersections, 2025. arXiv:2506.09019.

[CH98] Li Chiang and Yu-Ching Hung. On Hilbert-Kunz functions of some hypersurfaces. *Journal of Algebra*, 199(2):499–527, 1998. doi:10.1006/jabr.1997.7206.

[CR26] Nicholas Cox-Steib and Richard Redner. The linear algebra of the Han–Monsky representation ring. *The Electronic Journal of Linear Algebra*, 42:329–348, 2026.

[CSTZ24] Alessio Caminata, Samuel Shideler, Kevin Tucker, and Francesco Zerman. F-signature functions of diagonal hypersurfaces, 2024. URL: https://arxiv.org/abs/2403.12863, arXiv:2403.12863.

[Han92] Chungsim Han. *The Hilbert-Kunz Function of a Diagonal Hypersurface*. Brandeis University, 1992.

[HL04] Craig Huneke and Graham Leuschke. Two theorems about maximal Cohen–Macaulay modules. *Mathematische Annalen*, 324, September 2004. doi:10.1007/s00208-002-0343-3.

[HM93] Chungsim Han and Paul Monsky. Some surprising Hilbert-Kunz functions. *Mathematische Zeitschrift*, 214(1):119–135, 1993. doi:10.1007/BF02572395.

[Kun76] Ernst Kunz. On Noetherian Rings of Characteristic p. *American Journal of Mathematics*, 98(4):999–1013, 1976. doi:10.2307/2374038.

[Men26a] Cheng Meng. Hilbert-Kunz multiplicity of quadrics decreases, 2026. arXiv:2606.04346.

[Men26b] Cheng Meng. Limits of F-invariants and Riemann-Stieltjes integral, 2026. arXiv:2507.13898.

[Mon83] Paul Monsky. The Hilbert-Kunz Function. *Mathematische Annalen*, 263:43–50, 1983. doi:10.1007/BF01457082.

[Mon08] Paul Monsky. Rationality of Hilbert-Kunz multiplicities: A likely counterexample. *Michigan Mathematical Journal*, 57:605–613, 2008. doi:10.1307/mmj/1220879427.

[MT04] Paul Monsky and Pedro Teixeira. P-Fractals and power series–I. Some 2 variable results. *Journal of Algebra*, 280(2):505–536, 2004. doi:10.1016/j.jalgebra.2004.05.016.

[MT06] Paul Monsky and Pedro Teixeira. P-Fractals and power series—II.: Some applications to Hilbert–Kunz theory. *Journal of algebra*, 304(1):237–255, 2006. doi:10.1016/j.jalgebra.2005.07.033.

[PSSY26] Igor Pak, Boris Shapiro, Ilya Smirnov, and Ken-ichi Yoshida. Hilbert–Kunz multiplicity of quadrics via Ehrhart theory, 2026. arXiv:2508.17915.

[Yos09] Ken-ichi Yoshida. Small Hilbert-Kunz multiplicity and (A_1)-type singularity. *Proceedings of the 4th Japan-Vietnam Joint Seminar on Commutative Algebra by and for Young Mathematicians, Meiji University, Japan*, 2009.

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